

Referee package (DRAFT) – H-A0 to H-ANCHOR: A-zero uniqueness as a sign-decomposition theorem

TECT verification-first repository · claim B2-PROPA-HLAYER

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1. Purpose and scope

This referee document concerns the replacement of the H-A0 hypothesis by H-ANCHOR via a sign-decomposition theorem for $A = 0$ uniqueness and zero-at-gap. **Scope:** the free-energy derivative $F'_0(m)$ on the window. **Not claimed:** only the sign-decomposition and its discharge of H-A0's U, Z conditions.

2. The result

A -zero uniqueness and zero-at-gap (H-A0's U, Z) are derived as a sign-decomposition theorem: $F'_0(m) = \frac{1}{2}M'(m)g(m)$ with $M' < 0$ (L1), g strictly decreasing on the window (L2), and $g < 0$ beyond the gap (L3); hence F'_0 has a unique sign change, discharging H-A0. Explicitly, $g(m_{\text{gap}}) = 0$, $g'(m) < 0$ on the certified window, and $g(m) < 0$ for $m > m_{\text{gap}}$; the unique sign change of $F'_0(m)$ occurs at the anchored gap point because $g(m_{\text{gap}}) = 0$ and $M'(m) < 0$.

3. Structure of the argument

Factor the derivative as $F'_0 = \frac{1}{2}M'g$. L1 gives $M' < 0$ (dominated convergence, the G-A0-DUI lemma). L2 shows g strictly decreasing (closed form). L3 shows $g < 0$ beyond the gap. The product's sign is therefore determined, yielding a unique stationary point at $A = 0$ and the zero-at-gap property – exactly H-A0's U, Z , now theorems rather than hypotheses.

4. Numerical reproduction

The sign-decomposition script reproduces the $M' < 0$, g -decreasing and $g < 0$ facts and the unique sign change with asserts; run it from the bundle. The expected output is $M'(m) < 0$, $g'(m) < 0$, $g(m) < 0$ beyond the gap, and no second stationary point.

5. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) *Attack the scope.* The result is confined to the scope stated in Sec. 1; a referee may push that boundary, which is fair — the package makes no claim outside it, and the precise scope is in the footer.
- (β) *Attack the tier / over-claim.* No prose here is stronger than the registered grade (footer no-overclaim line); the status is REVIEW DRAFT, NOT operator-confirmed, and the claim-ledger tier is the arbiter.

- (γ) *Attack reproduction / self-containment.* Each reproduction script carries self-test asserts and ships in the folder's bundle with its transitive dependencies (imports and runtime file reads, resolver v1.3.0); a referee runs them and diffs against `expected/`.

Quantitative sanity check (mandatory): the falsifier (footer) is concrete and pre-registered, so the claim is refutable by a single explicit construction; and the numerical values it cites are reproduced by the folder's scripts (all asserts pass).

6. Result footer

Result ID:	B2-PROPA-HLAYER / H-A0-removal referee package (DRAFT)
Precise statement:	A-zero uniqueness + zero-at-gap (H-A0's U,Z) derived as a sign-decomposition theorem: $F_0'(m) = (1/2)M'(m)g(m)$ with $M' < 0$ (L1), g strictly decreasing on the window (L2), $g < 0$ beyond the gap (L3); unique sign change discharges H-A0.
Scope:	the free-energy derivative $F_0'(m)$ on the window; H-A0 discharge.
Evidence grade:	sign-decomposition theorem (T6/T7); EXECUTED.
Expected output:	$M'(m) < 0$; $g'(m) < 0$; $g(m_{\text{gap}}) = 0$; $g(m) < 0$ beyond the gap; no second stationary point.
Reproduction command:	python codes/vacuum/ha0_sign_decomposition.py
Falsifier:	a window point violating $M' < 0$ or g -monotonicity; or a second stationary point.
Tier before / after:	DRAFT referee package; H-A0 discharged to H-ANCHOR. AUXILIARY DRAFT ACCEPTED (2026-06-11); not promoted to published.
No-overclaim statement:	H-A0's U,Z become theorems on the window; not the full Prop-A comparison.
Next required action:	remain an auxiliary cited lemma (DRAFT bundle only); not promoted to a published package.